

TarAIntino: AI Movie Generation Infrastructure

An End-to-End System for Personalized Cinematic Creation

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Abstract

We propose TarAIntino, an end-to-end system to generate personalized movie trailers (and eventually full movies) using AI. Our project addresses a critical gap in the generative AI landscape: while model capabilities are advancing rapidly, the infrastructure to make them accessible and intuitive for creative tasks is lagging. We are not building a new video model; we are building the essential "glue" that connects user preferences, narrative logic, and generative APIs into a seamless, usable application. Our core innovations lie in applying advanced preference elicitation techniques to map a user's abstract tastes onto a continuous "cinematic taste space" and orchestrating a resilient, production-grade pipeline to translate that taste into a coherent visual story.

1. Background and Motivation

The entertainment industry is on the cusp of a paradigm shift. Visionaries like Marvel director Joe Russo predict AI could "create a fully fledged movie" within years [1], while industry experts forecast that over 90% of online content may be AI-generated by 2026 [2]. The technical foundation is rapidly maturing, with text-to-video models from Runway, Google, and Luma achieving impressive quality.

However, a fundamental challenge remains: *usability*. The best generative model is ineffective if the user cannot intuitively translate their creative intent into a series of precise prompts. Most people will not invest more effort into "programming" a movie than simply watching one. Our motivation is to solve this infrastructure problem. We believe specifying a dream movie should be as easy as a Tinder swipe and not a complex technical task.

2. Problem Statement

To build a modular, end-to-end infrastructure that takes a user's abstract cinematic preferences, elicited through an adaptive quiz, and automatically produces a short, coherent, AI-generated movie trailer reflecting those tastes, without requiring technical knowledge from the user.

3. Data Sources

Our system's intelligence is built upon a foundation of rich, well-structured film data. Data availability is confirmed, and all sources are appropriate for an academic project.

- **Primary Source: MovieLens 25M & Tag Genome** [3]. This dataset is the cornerstone of our project. It contains 25 million ratings, but critically, it includes the **Tag Genome**: over 15 million continuous relevance scores (0.0 to 1.0) for 1,129 descriptive tags (e.g., "noir," "dystopian," "whimsical") across 62,000 films. This allows us to move beyond discrete genres and model taste in a high-dimensional vector space.

- **Supporting Source: IMDb Datasets**. Provides comprehensive metadata (cast, crew, plot summaries) for enriching our film database under a non-commercial use license.

- **Data Quality Concerns**: The primary challenge is merging MovieLens and IMDb data, which use different ID schemes. We will mitigate this by using provided mapping files and implementing a fallback based on fuzzy title matching.

4. Technical Approach and Design

Our architecture is designed as a production-grade, orchestrated workflow. Each stage is a distinct, modular service, ensuring scalability and resilience.

4.1 Innovation 1: Continuous Taste Space

Traditional recommenders use discrete genres. We use the Tag Genome to create a continuous **Taste Vector** $\theta \in \mathbb{R}^d$ (where $d \approx 400$) for each user. A film is no longer just "Action"; it's a point in this space (e.g., *noir*=0.8, *romantic*=0.3, *violent*=0.5). This enables nuance like finding a cinematic midpoint between *Blade Runner* and *Inception*, by performing vector arithmetic. This approach is grounded in the foundational work of Vig et al. [4].

4.2 Innovation 2: Adaptive Preference Elicitation

To populate the Taste Vector, we replace static surveys with a dynamic, "Akinator-style" quiz. We will apply **Item Response Theory (IRT)** [5], a technique from adaptive testing, to select questions that maximally reduce uncertainty about the user's latent taste vector. Each question (e.g., a pairwise comparison modeled via the Bradley-Terry model [6]) is chosen to provide the most information, allowing us to build a rich profile in just 10-15 interactions. In case of insufficient response from the user end, using **Singular Value Decomposition (SVD)** to infer the user preferences might be a good approach.

4.3 System Architecture

The entire process is managed by a workflow orchestrator (à la Temporal.io [7]) to handle state, retries, and asynchronous execution of long-running video generation tasks.

5. Scope and Project Objectives

Our goal is to create a fully automated, PoC pipeline that is robust and demonstrates key MLOps principles.

5.1 In-Scope Objectives

1. Implement an adaptive quiz engine to generate a user taste vector.
2. Develop an LLM-based agent to produce a structured `MovieSpec.json`.

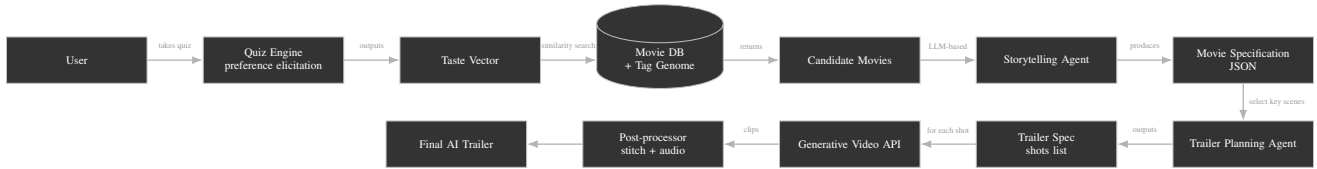


Figure 1: High level workflow in two stages: (1) Preference Elicitation and Story Definition (top row), and (2) Trailer Planning and Video Production (bottom row). This modular pipeline ensures a robust process from user input to final trailer.

3. Develop an LLM-based agent to produce a structured `TrailerSpec.json` based on `MovieSpec.json`.
4. Convert the trailer spec into a shot list and generate video clips via at least one external API.
5. Automate the stitching of clips and addition of AI-generated audio into a final trailer file.
6. Containerize the application components to ensure reproducibility.

5.2 Out-of-Scope

We will *not* train our own video generation models. We will also *not* focus on building a multi-user frontend UI; a simple interface is sufficient for demonstration.

5.3 Alignment with Course Components

This project is explicitly designed to engage with core MLOps challenges:

- **Large Data:** We will process the MovieLens dataset with its 15 million tag relevance scores.
- **Complex Models:** We integrate a pipeline of diverse models: IRT for preference and LLMs and Diffusion models API for narrative video.
- **Scalability:** Our architecture, designed around a workflow orchestrator, is inherently scalable. Video generation can be parallelized as asynchronous jobs.

6. Limitations and Risk Mitigation

We have identified key risks and have proactive mitigation strategies in place:

- **Video Quality & Consistency:** AI-generated characters lack visual consistency across shots. So we will design trailers to play to AI's strengths: use silhouettes, dynamic montages, wide shots, and non-photorealistic styles. We will also explore emerging techniques like reference-based generation [8].
- **High API Costs:** Video generation is expensive (\$50-100 per trailer is possible). So we will implement caching for prompts and generated shots. Budget API credits carefully. Develop a "storyboard mode" (still images) as a low-cost fallback for testing the pipeline.
- **Pipeline Complexity:** Integrating multiple disparate services is prone to failure. So we will use a workflow orchestrator (Temporal) to manage state and provide automatic retries. Enforce strict schemas (JSONSchema) for data transfer between modules.

7. Fun Factor

Watching AI-generated trailers based on our own tastes makes the project playful and engaging, sparking fun discussions about what we like (or don't) in the results.

8. Application Mock Design

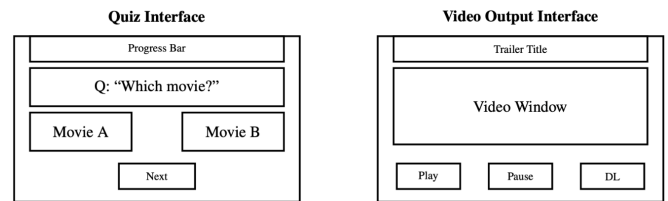


Figure 2: Wireframes of two key user interfaces: (Left) Adaptive Quiz for preference elicitation, (Right) Video output interface with player, controls, and metadata.

9. Milestones

- **Milestone 1 (Sep 25):** Project proposal and data validation. (*Completed*)
- **Milestone 2 (Mid-Oct):** Data processing, pipeline set up and baseline model. Can generate trailer scene prompts based on `MovieSpec`.
- **Milestone 3 (Early-Nov):** Work on front-end development and adaptive quiz engine with taste vector.
- **Milestone 4 (Late-Nov):** MVP application. Dynamic quiz is functional, LLM generates a story, and the pipeline produces a short video trailer.
- **Milestone 4 (Dec):** Final report delivery. Polished trailer output, full system demonstration.

References

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